

GOVERNMENT OF SIERRA LEONE
MINISTRY OF HEALTH AND SANITATION
NATIONAL MALARIA CONTROL PROGRAMME

Guidelines for Case Management of Malaria

SIXTH EDITION 2024

EXECUTIVE SUMMARY

In Sierra Leone, malaria remains a major health and socio-economic challenge, as emphasized in the National Health Sector Strategic Plan (NHSSP 2021-2025). Recognizing the burden of the disease, the government has prioritized malaria control, focusing on early diagnosis and effective treatment as cornerstones of its strategy to improve public health outcomes.

To guide healthcare professionals, this national malaria case management guidelines outline best practices for treating the disease at all levels of the healthcare system in the country. A key recommendation is that all suspected malaria cases undergo parasitological confirmation through microscopy or rapid diagnostic tests (RDTs) before initiating treatment. This approach ensures accurate diagnosis and appropriate care, helping to prevent the misuse of antimalarial drugs and combat resistance.

For uncomplicated malaria, the first-line treatment is Artemether-Lumefantrine (AL). Artesunate-Amodiaquine (AS+AQ) is recommended as an alternative when AL is unavailable. In cases of severe malaria, parenteral artesunate is the treatment of choice for initiating treatment due to its rapid efficacy in reducing malaria complications. In situations where parenteral artesunate is not immediately feasible, other options such as intramuscular artesunate and rectal artesunate serve as a pre-referral treatment to stabilize the patient while facilitating transfer to a facility capable of administering the full treatment course for severe malaria.

These guidelines ensure improved quality of care by standardizing malaria management practices across Sierra Leone. They emphasize prompt treatment and align with global recommendations by the WHO, ensuring that treatments remain effective while preventing resistance. This comprehensive approach underscores the country's commitment to improving malaria outcomes and aligns with global best practices recommended by the World Health Organization (WHO). By ensuring access to timely and effective treatment, Sierra Leone aims to mitigate the impact of malaria and achieve significant health sector gains.

FOREWORD

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LIST OF ACRONYMS

ACT – Artemisinin-based Combination Therapy
ADR – Adverse Drug Reaction
AL – Artemether-Lumefantrine
AQ – Amodiaquine
AS – Artesunate
ASAQ – Artesunate-Amodiaquine
CHA – Community Health Assistant
CHW – Community Health Worker
CCMm – Community Case Management of Malaria
DHS – Demographic and Health Survey
DHIS – District Health Information System
DIC – Disseminated Intravascular Coagulation
DP – Dihydroartemisinin-Piperaquine
ETAT+ – Emergency Triage, Assessment, and Treatment Plus
Hb – Hemoglobin
HF – Health Facility
ICU – Intensive Care Unit
IM – Intramuscular
IPTp – Intermittent Preventive Treatment in Pregnancy
IRS – Indoor Residual Spraying
ITN – Insecticide-Treated Net
LLIN – Long-Lasting Insecticidal Net
mRDT – Malaria Rapid Diagnostic Test
MCAT – Multi-Country Assignment Team
MDA – Mass Drug Administration
MICS – Multiple Indicator Cluster Survey
MIS – Malaria Indicator Survey
MIP – Malaria in Pregnancy
MoH – Ministry of Health

M&E – Monitoring and Evaluation
NMCP – National Malaria Control Program
NHSSP – National Health Sector Strategic Plan
NTDs – Neglected Tropical Diseases
ORS – Oral Rehydration Solution
PCV – Packed Cell Volume
PfHRP2 – Plasmodium falciparum Histidine-Rich Protein-2
PfK13 – Plasmodium falciparum Kelch 13
PMC – Perennial Malaria Chemoprevention
RDT – Rapid Diagnostic Test
SBC – Social and Behaviour Change
SP – Sulphadoxine-Pyrimethamine
TET – Therapeutic Efficacy Testing
WHO – World Health Organization

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communities, often in challenging conditions, is the cornerstone of malaria control efforts in Sierra Leone.

This document is a testament to the collective effort of all those working towards a malaria-free Sierra Leone.

CHAPTER 1 INTRODUCTION

Malaria continues to pose a significant health challenge in Sierra Leone and is endemic across the country. There is a stable year-round transmission that peaks during the rainy seasons in May and October. Children under five and pregnant women remain the most vulnerable, experiencing disproportionately high rates of morbidity and mortality.

Data from Malaria Indicator Surveys indicate positive trends. Malaria prevalence among children under five dropped from 40% in 2016 to 22% in 2021, underscoring the effectiveness of control efforts. However, the annual parasite incidence saw a slight rise, signaling ongoing challenges in sustaining the gains achieved. Enhanced measures, including digitized ITN distribution campaigns, aim to address these gaps by improving usage and coverage.

In Sierra Leone, malaria is caused primarily by *Plasmodium falciparum*, which is responsible for over 90% of malaria cases. Other malaria-causing species, such as *Plasmodium vivax*, *Plasmodium malariae*, and *Plasmodium ovale*, are less common in the country. The high prevalence of *P. falciparum* is driven by favorable conditions for transmission, including a tropical climate and year-round mosquito breeding supported by swamps, rice paddies, and coastal mangroves. These areas serve as breeding grounds for the primary malaria vectors: *Anopheles gambiae*, *Anopheles funestus*, and *Anopheles melas*.

The 2021–2025 National Malaria Elimination Strategic Plan reflects Sierra Leone’s commitment to achieving a malaria-free future. With a goal of reducing malaria morbidity and mortality by 75% by 2025 compared to 2015 levels, the country is focusing on integrating new tools like the vaccine and optimizing established interventions. The introduction of the RTS, S malaria vaccine in 2024 marked an important step in Sierra Leone’s fight against the disease. Targeted at children aged 6 to 23 months, the vaccine complements existing measures like the widespread distribution of insecticide-treated nets (ITNs), indoor residual spraying (IRS), and access to prompt diagnostic and treatment services. This integrated strategy aims to reduce the incidence and mortality associated with malaria, especially among the most affected populations.

1.1 MALARIA DRUG RESISTANCE PATTERN IN SIERRA LEONE

The current situation regarding malaria treatment in Sierra Leone indicates significant progress in maintaining the efficacy of antimalarial drugs. Recent therapeutic efficacy studies (TES) in the country confirm that artemisinin-based combination therapies (ACTs) such as artesunate-amodiaquine (ASAQ), artemether-lumefantrine (AL), and dihydroartemisinin-piperaquine (DP) remain highly effective for treating uncomplicated *Plasmodium falciparum* malaria. These studies report no cases of resistance-associated mutations, such as those in the Pfk13 gene linked to artemisinin resistance or the Pfplasmepsin2 gene associated with piperaquine resistance. Adjusted cure rates (ACPR) after molecular corrections consistently reach 100%, reflecting the continued potency of these treatments.

Additionally, there is minimal presence of antifolate-resistant markers, such as the quintuple mutant associated with sulfadoxine-pyrimethamine (SP) resistance. This supports the strategic decision of the National Malaria Control Programme (NMCP) to implement perennial malaria

chemoprophylaxis SP in districts with high malaria transmission. However, vigilance remains critical due to the emerging risks of antimalarial resistance across Africa, driven by mutations and regional genetic markers. Continuous surveillance using both therapeutic efficacy studies and molecular analyses is a cornerstone of Sierra Leone's malaria program, aligned with WHO recommendations for monitoring drug resistance trends.

1.2 GENERAL OBJECTIVE

The main objective of this document is to provide updated recommendations and regulations for the care of patients with malaria, which is aligned with the latest WHO guidelines on Malaria (October 2023), and current evidence-based best practices in malaria case management.

1.3 SPECIFIC OBJECTIVE

To guide stakeholders across all levels of the healthcare system in managing malaria cases effectively:

- Ensure a prompt and accurate diagnosis of malaria.
- Provide timely and appropriate treatment.
- Early recognition of the danger signs of severe malaria.
- Prompt referral of cases to the appropriate level equipped to provide adequate support services and care.
- Build capacity and equip all levels of health service delivery, including community health systems, to implement Community Case Management of malaria (CCMm).
- Recognize the importance of full compliance with treatment schedules.

1.4 TARGET GROUP

The guidelines are designed for all stakeholders involved in malaria control, including health professionals in both public and private sectors, non-governmental organizations, research and training institutions, the pharmaceutical industry, and other partners contributing to health or malaria control efforts

1.5 HEALTH CARE LEVELS

Health service delivery within the Ministry of Health and Sanitation in Sierra Leone is organized into three levels. These levels are defined based on the population served, the competencies of healthcare providers, the availability of equipment, and the volume and quality of support services, as outlined in the 2015 Basic Package of Essential Health Services for Sierra Leone.

- I. **Primary Level:** Includes Peripheral Health Units, Community Health Centers, Community Health Posts, Maternal and Child Health Posts, and Community Health Workers. At this level, only outpatient services for malaria are available.
- II. **Secondary Level:** Comprises district hospitals and mission hospitals, staffed by medical officers and nursing personnel, with laboratories available for diagnosing and monitoring malaria cases. Both outpatient and in-patient services are available at this level.
- III. **Tertiary Level:** Encompasses larger district hospitals and National Referral Hospitals, which provide specialized and advanced healthcare services. This level also offers both in-patient and out-patient malaria services

CHAPTER 2 CLASSIFICATION OF MALARIA

Malaria symptoms can resemble other illnesses, making parasitological testing essential for confirmation. Fever is the most common symptom, with an incubation period of 7 to 21 days for *Plasmodium falciparum*. If treated promptly, the disease is usually mild, but if left untreated, it can lead to severe complications due to the sequestration of infected red blood cells in deep tissues. In pregnant women, *P. falciparum* malaria poses serious risks to both mother and fetus, including placental infection, maternal anemia, and poor birth outcomes.

1.1 CLASSIFICATION

Malaria is classified into two main categories:

- **Uncomplicated Malaria:** Uncomplicated malaria is characterized by the presence of fever or a recent history of fever (within the last 2-3 days), accompanied by a confirmed parasitological diagnosis (positive malaria RDT or malaria parasites observed through microscopy). This occurs in the absence of clinical or laboratory signs of severe disease or vital organ dysfunction.
- **Severe Malaria:** Severe malaria is characterized by the presence of fever or a recent history of fever, along with one or more clinical or laboratory signs of severe disease or vital organ dysfunction. This is accompanied by a confirmed parasitological diagnosis, either through a positive RDT or microscopy.

The clinical and laboratory features of severe malaria and vital organ dysfunction are detailed in Chapter 4 below.

CHAPTER 3 UNCOMPLICATED MALARIA

Uncomplicated malaria is a symptomatic infection with malaria parasitaemia, without signs of severity or evidence (clinical or laboratory) of vital organ dysfunction. Infection with malaria parasites can cause a range of non-specific symptoms, with fever being the most common. Headaches, joint aches, and general discomfort typically accompany the fever. In infants and young children, the onset of malaria symptoms may resemble an influenza-like illness, with early signs including poor appetite, restlessness, cough, diarrhea, and a loss of normal interest in their surroundings.

2.1 DEFINITION OF UNCOMPLICATED MALARIA

Uncomplicated malaria is characterized by symptoms such as fever, chills, and headache, confirmed through a positive malaria test without signs of severe disease.

2.2 SIGNS AND SYMPTOMS

The signs and symptoms of uncomplicated malaria are often non-specific, and no single combination of clinical features reliably distinguishes malaria from other causes of fever. As a result, the clinical presentation may closely resemble other illnesses, including pneumonia, meningitis, typhoid fever, Lassa fever, yellow fever, COVID-19, and others.

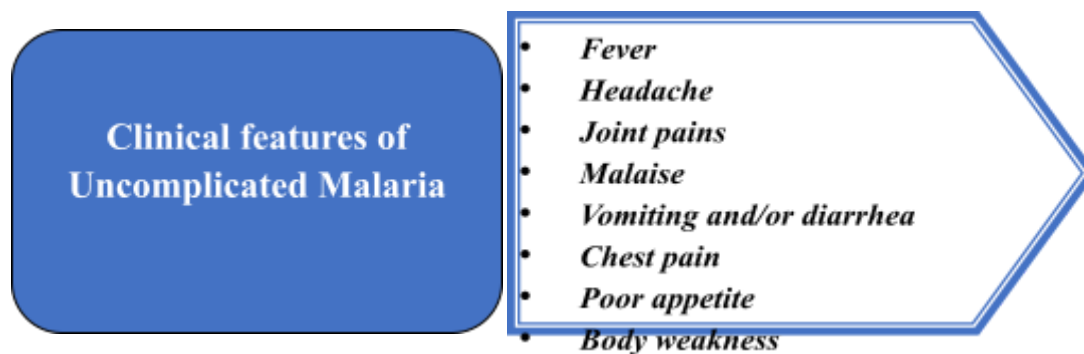


Figure SEQ Figure * ARABIC 1: Clinical features of uncomplicated malaria

Patients suffering from uncomplicated malaria commonly complain of:

- Fever (37.5°C or higher); a history of fever for 2-3 days
- Chills or feeling unusually cold
- Rigors or shivering
- Headache

Other features include:

- Generalized body and joint pains
- Nausea
- Vomiting
- Diarrhea
- Loss of appetite
- Sweating
- Pallor
- Palpable liver and spleen

2.3 DIAGNOSIS

RECOMMENDATION: All cases of suspected malaria should be confirmed with a parasitological test (with either microscopy or mRDT)

A person presenting with a recent history of fever (within the past 2-3 days) or a peripheral temperature of $\geq 37.5^{\circ}\text{C}$ on examination can be classified as a suspected case of malaria.

Parasitological confirmation of malaria should be part of good clinical practice to ensure the provision of high-quality care to patients. Definitive diagnosis of malaria should be made by parasitological confirmation either through microscopy, which detects malaria parasites in blood smears (see Annex 1 for details on reporting blood smear results), or through the detection of parasite antigens using malaria rapid diagnostic tests (mRDTs). mRDTs can also be used at all levels of the health care delivery system, including at the community level by Community Health Workers to confirm diagnosis of malaria.

Test results should be made available within two hours of specimen collection. Both microscopy and mRDTs must be supported by a robust quality assurance program to maintain diagnostic accuracy and reliability. Antimalarial treatment should be administered **ONLY** to patients with confirmed positive test results. In cases where the malaria test is negative, patients should be reassessed for other common causes of fever and treated accordingly.

2.4 TREATMENT

RECOMMENDATION: Treat all case of uncomplicated malaria with Artemether+Lumefantrine (AL). Artesunate+Amodiaquine (ASAQ) can be used as a second-line treatment.

2.4.1 TREATMENT OBJECTIVES

The **clinical objectives** of treating uncomplicated malaria are to:

- To cure the infection as rapidly as possible (cure is defined as the elimination from the body of the parasite that causes the illness).
- To prevent the progression of uncomplicated malaria to severe disease, and additional morbidity associated with treatment failure.

The public health objectives of treatment are to:

- Prevent onward transmission
- Prevent the emergence and spread of resistance to antimalarial medicines

2.4.2 TREATMENT RECOMMENDATIONS

Confirmed malaria cases should receive prompt and appropriate treatment to ensure effective management and prevent complications. Treatment based solely on clinical suspicion should be considered only when parasitological diagnosis is unavailable or infeasible. The recommended first-line treatment for uncomplicated malaria should remain consistent across all levels of the healthcare delivery system.

The **MEDICINE OF CHOICE** for the treatment of all cases of uncomplicated malaria, including in pregnant women across all trimesters, is **Artemether-Lumefantrine (AL)**.

In situations where AL is unavailable or cannot be tolerated, the recommended **second-line treatment** for uncomplicated malaria is **Artesunate-Amodiaquine (ASAQ)**.

Table 1: Recommended dosage schedule for Artemether-Lumefantrine treatment (20/120mg)

Weight (Kg)	Age	20/120mg					
		Day 1		Day 2		Day 3	
		0 hours (Initial dose)	8 hours later	Morning	Evening	Morning	Evening
< 14	>3yrs	1 tab/dose	1 tab/dose	1 tab/dose	1 tab/dose	1 tab/dose	1 tab/dose

15-24	4-8yrs	2 tabs/dose	2 tabs/dose	2 tabs/dose	2 tabs/dose	2 tabs/dose	2 tabs/dose
25-34	9-14yrs	3 tabs/dose	3 tabs/dose	3 tabs/dose	3 tabs/dose	3 tabs/dose	3 tabs/dose
>35	>14yrs	4 tab/dose	4 tab/dose	4 tab/dose	4 tab/dose	4 tab/dose	4 tabs/dose

Table 2: Recommended dosage schedule for Artemether-Lumefantrine treatment (80/480mg)

Weight (Kg)	Age	80/480mg					
		Day 1		Day 2		Day 3	
		0 hours (initial dose)	8 hours later	Morning	Evening	Morning	Evening
> 35	>14yrs	1 tab/dose	1 tab/dose	1 tab/dose	1 tab/dose	1 tab/dose	1 tab/dose

Note: Absorption of lumefantrine is enhanced by co-administration with a meal. It is therefore recommended that this ACT be taken immediately after food or a fat-containing drink (e.g., milk), particularly on the second and third days of treatment.

Table 3: Recommended dosage schedule for fixed combination of ASAQ treatment

Weight (Kg)	Age	Dosage
< 9	Less than 11 months	1 tablet (25mg artesunate/67.5 mg amodiaquine) per day for 3 days
≥9 to <18	1 to 5 years	1 tablet (50mg artesunate/135 mg amodiaquine) per day for 3 days
≥18 to <36	6 to 13 years	1 tablet (100mg artesunate/270 mg amodiaquine) per day for 3 days
≥ 36	14 years and above	2 tablets (100mg artesunate/270 mg amodiaquine) per day for 3 days

Note: Weight, if available, should be used as the first indicator to determine dosage. If weight is unavailable (e.g., no access to a scale), age can be used as a second indicator to determine dosage

Patients should be advised to seek immediate medical attention if symptoms worsen or if any of the following danger signs develop:

- Convulsions
- Lethargy or unconsciousness
- Excessive sleepiness or drowsiness
- Abnormal breathing

- Protracted vomiting
- Inability to eat, drink, or breastfeed
- Lack of response to treatment 48 hours after starting medication

The first dose of treatment should be administered in the clinic or by the Community Health Worker under supervision. If vomiting occurs within 30 minutes of administration, the dose should be repeated. The patient should be provided with the second and third doses to take home and be counseled on the proper administration of the medication, as well as the necessity of returning to the health facility if vomiting persists.

If vomiting persists, this is an indication for **REFERRAL**. Administer pre-referral (see page 27) before transferring the patient to an appropriate facility.

2.4.3 SUPPORTIVE TREATMENT

Supportive treatment is an important component of malaria case management.

- If the patient presents with an axillary temperature above 37.5°C or feels warm to the touch during examination, administer an antipyretic, preferably paracetamol, particularly for children.
- Remove excess clothing in children with fever.
- Advise mothers and caregivers to provide extra fluids, such as breast milk, drinking water, diluted fruit juices, coconut water, or oral rehydration solution (ORS), to keep the patient hydrated.
- Encourage continued feeding during illness to maintain nutrient intake.
- For pregnant women, encourage adherence to the protocol for taking iron and folic acid tablets to support maternal and fetal health.

Paracetamol tablets should be administered every 6 hours until the fever subsides. Weight, if available, should be used as the first indicator to determine dosage. If weight is unavailable (e.g., no access to a scale), age can be used as a second indicator to determine dosage

Table 4: Dosage schedule for Paracetamol tablets (500mg)

Weight (Kg)	Age	Dosage
4 to 10	2 to 11 months	¼ tablet
10 to 18	1 to 5 years	½ tablet
18 to 35	6 to 9 years	1 tablet
35 to 45	10 to 14 years	1½ tablets
>45	15+ years	2 tablets

Table 5: Dosage schedule for Paracetamol Syrup (125mg/5ml)

Age	Dosage
0 to 6 months	2.5mls
7 to 11 months	5mls
1 to 2 years	7.5mls
2 to 4 years	10mls

Note: Paracetamol syrup (125 mg/5mls) should be given at a dose of 10mg/kg every 6 hours until the fever subsides. A large challenge with syrup is ensuring the correct dosage. As such, dispersible tablets are much preferred to syrups.

Table 6: Dosage schedule for Paracetamol tablets (100mg)

Age	Dosage
0 to 6 months	½ tablet
7 to 11 months	1 tablet
1 to 2 years	1 ½ tablets
2 to 4 years	2 tablets
5 to 9 years	3 tablets

Advise mothers and caregivers to seek immediate medical attention if the patient's condition worsens. Additionally, they should be encouraged to report any suspected adverse events associated with the medication for appropriate follow-up and management.

2.5 TREATMENT OF MALARIA CAUSED BY OTHER SPECIES

Malaria caused by *Plasmodium ovale*, *Plasmodium malariae*, and *Plasmodium vivax* are also susceptible to artemether-lumefantrine (AL). Therefore, patients infected with these species—either in isolation or in combination with *P. falciparum*—should be treated with AL following the same regimen as for *P. falciparum* malaria. However, in cases of *P. vivax* and *P. ovale* infections, a seven-day course of primaquine should be added to prevent relapse, provided there are no contraindications such as Glucose-6-Phosphate Dehydrogenase (G6PD) deficiency.

Table 7: Recommended Daily Dosing Regimen for *P. ovale* and *P. vivax* radical cure using Primaquine 3.75 mg or 7.5 mg base

Weight (Kg)	Primaquine tablet (3.75 mg base)	Primaquine tablet (7.5 mg base)
<10 (>6months)	1 tablet	½ tablet

10 to < 15	2 tablets	1 tablet
15 to < 30	4 tablets	2 tablets
30 to < 60	6 tablets	3 tablets
>60	8 tablets	4 tablets

2.6 RECURRENT P. FALCIPARUM MALARIA

Recurrence of *Plasmodium falciparum* malaria can occur due to either re-infection or recrudescence (treatment failure).

Treatment failure occurring within 28 days after initial treatment may result from drug resistance or inadequate exposure to the drug. Contributing factors include sub-optimal dosing, poor adherence, vomiting, individual variations in pharmacokinetics, or the use of substandard medicines. It is crucial to assess the patient's history to determine if they vomited the previous treatment or failed to complete the full course. When possible, treatment failure should be confirmed parasitologically using microscopy or Lactate Dehydrogenase (LDH)-based rapid diagnostic tests (RDTs), as *Plasmodium falciparum* histidine-rich protein-2 (PfHRP2)-based RDTs may remain positive for weeks following the initial infection, even without recrudescence. If microscopy or LDH-based mRDTs are unavailable, the patient should be referred to a secondary level of care.

In the case of treatment failure, the second-line ACT, artesunate-amodiaquine (ASAQ), should be administered. If the initial treatment was with ASAQ, then artemether-lumefantrine (AL) should be used for re-treatment.

If fever and parasitaemia reappear more than 28 days after initial treatment, it may be due to either recrudescence or a new infection. The distinction between these two causes can only be made through PCR genotyping of parasites from both the initial and recurrent infections. Since PCR is not routinely employed in patient management, any presumed treatment failures occurring more than 28 days after the initial treatment should, from an operational perspective, be considered as new infections and managed with the first-line ACT (AL).

CHAPTER 4 SEVERE MALARIA

Severe Malaria is a Medical Emergency!!!

Severe malaria is a multi-syndromic disease characterized by clinical or laboratory evidence of vital organ dysfunction due to malaria infection. While it can affect individuals of all ages, the risk is particularly high in children under five years of age, pregnant women, and those with low or no malaria immunity. Delayed diagnosis and inappropriate treatment of uncomplicated malaria, particularly in infants and children, can rapidly progress to severe malaria. Severe malaria is primarily caused by *Plasmodium falciparum* infection. It is, therefore, crucial to confirm the diagnosis of severe malaria through the detection of asexual forms of *Plasmodium falciparum* in the blood.

3.1 DEFINITION OF SEVERE MALARIA

Severe malaria is defined as *Plasmodium falciparum* infection accompanied by any of the following life-threatening conditions listed in Fig 2

3.2 SIGNS AND SYMPTOMS

Malaria is considered severe if a patient presents with any one or a combination of the following clinical manifestations and/or laboratory findings that indicate severe disease:

Clinical features of severe malaria:

- Prostration (inability to feed, sit, stand, or walk without assistance due to extreme weakness)
- Altered consciousness (change in behavior, confusion, delirium, or coma lasting more than 30 minutes after convulsion)
- Multiple convulsions (2 or more episodes within 24 hours)
- Respiratory distress (Deep fast breathing resulting from acidosis) and or pulmonary oedema (with/without basal crackles visible on radiology)
- Spontaneous bleeding or disseminated intravascular coagulation (DIC)
- Shock or circulatory collapse (cold limbs, weak rapid pulse)
- Repeated vomiting or inability to tolerate oral feeding
- Clinical jaundice (yellow coloration of the eyes) plus evidence of other vital organ dysfunction
- Dark of cola-colored urine (sign of intravascular haemolysis)
- Oliguria (scanty urine) or anuria (no urine), suggestive of kidney injury
- Severe pallor (as a result of severe anemia)
- Symptoms of hypoglycemia (sweating, pupillary dilation, abnormal breathing, coldness, prostration, or altered sensorium)

Laboratory findings in severe malaria:

- Severe anemia (Hb <5 g/dL or PCV <15% in children; Hb <7 g/dL or PCV <20% in adults)
- Hypoglycemia (blood glucose <2.2 mmol/L or <40 mg/dL)
- Acute kidney injury (Serum creatinine >265 μ mol/L or a 1.5-fold increase from baseline within 7 days or an absolute increase of 26.5 μ mol/L from baseline within 48 hours or Serum Urea > 20 mmol/L)
- Jaundice (Serum bilirubin >50 μ mol/L (3 mg/dL)
- Metabolic Acidosis (plasma bicarbonate < 15 mmol/L or base deficit of > 8 mmol/L)
- Acute pulmonary edema or difficulty in breathing (Adult respiratory distress syndrome)
- Hyperparasitemia (>500,000 parasites/ μ L or >10% RBCs infected)
- Hyperlactaemia (Serum/Plasma Lactate > 5 mmol/L)
- Pulmonary Oedema (Radiological confirmation)

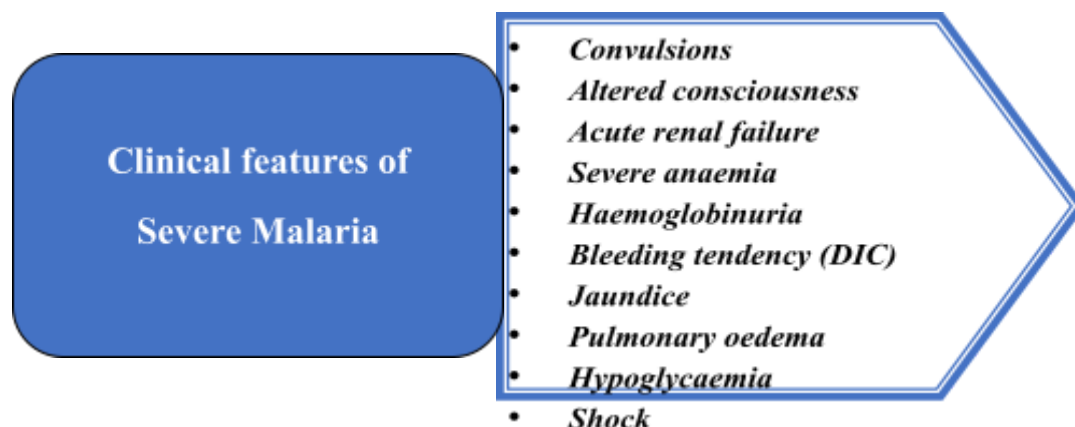


Figure SEQ Figure * ARABIC 2: Clinical features of severe malaria

The most common complications of severe malaria, which are responsible for the majority of deaths in children under five, include:

- Cerebral malaria – Prolonged coma not attributed to any other cause in a patient with falciparum malaria.
- Severe malaria anaemia
- Respiratory distress (metabolic acidosis, pulmonary oedema)
- Circulatory collapse (shock, severe dehydration, sepsis, etc.,)
- Hypoglycemia
- Hyperlactataemia

Children with fever suspected of having severe malaria should receive immediate treatment, while parasitological diagnosis with microscopy or mRDT is performed. In addition, they should also concurrently be examined and investigated for other causes of fever (such as respiratory infection, sepsis, meningitis, Lassa fever, etc.) and appropriately managed while awaiting laboratory results.

3.3 CLINICAL AND PARASITOLOGICAL DIAGNOSIS

The diagnosis of severe malaria, like uncomplicated malaria, is based on a thorough history, physical examination, and laboratory investigations. This includes confirmation through a malaria diagnostic test, either microscopy or mRDT.

The clinical diagnosis of severe malaria should be made in any patient presenting with the following:

- **Fever** (peripheral temperature $\geq 37.5^{\circ}\text{C}$) or a recent history of fever, AND
- **A positive malaria diagnostic test**, AND
- Any sign or symptom of severe disease or vital organ dysfunction, as outlined in Fig 2 above.

In children under 5 years of age, a clinical diagnosis of severe malaria can be made if the following criteria are met:

- **Fever** (peripheral temperature $\geq 37.5^{\circ}\text{C}$) or a recent history of fever, AND
- **A positive malaria diagnostic test**, AND
- Any “**general danger sign in young children**” from the list shown in Fig 3 below. The term “general danger sign” is a critical concept in the Integrated Management of Neonatal and Childhood Illnesses (IMNCI). These signs are non-specific clinical indicators that may signal a serious underlying illness requiring immediate medical attention.

General danger signs in young children

- *Vomiting everything*
- *Inability to drink or breastfeed*
- *Extreme weakness (prostration)*
- *Convulsions*

Figure SEQ Figure * ARABIC 3: General danger signs in young children

These criteria help identify patients who require immediate and intensive treatment to manage severe malaria effectively. Timely referral to a healthcare facility is crucial for proper management and to prevent complications.

Common errors in diagnosis:

- Failure to adequately identify features of severity and complications
- Failure to diagnose other associated infections and comorbidities

3.4 MANAGEMENT OF SEVERE MALARIA

Given that severe malaria is a medical emergency, it requires prompt clinical assessment and intervention. If the healthcare facility is unable to manage the case, the patient should be referred to a higher-level facility. For critically ill patients, particularly those in a coma, care should be provided in a special observation unit or intensive care unit (ICU).

It is recommended that the management of severe malaria is undertaken using the below guidance:

- I. Clinical Assessment: to determine the severity of the illness.
- II. Commence initial Emergency Treatment to stabilize the patient and address any life-threatening complications.
- III. Administer specific Antimalarial Treatment based on the patient's condition.
- IV. Adjunctive Therapy and Supportive Care as required.
- V. Assessment of recovery and follow up

3.4.1 TREATMENT OBJECTIVES

- To prevent the patient from dying.
- To prevent disabilities.
- To restore previous physical and mental state.

3.4.2 INITIAL EMERGENCY TREATMENT

The initial management of severe malaria should be done as follows:

- **Clear and Maintain Airway:** Ensure the patient's airway is open and clear.
- **Positioning:** If the patient is comatose, position them semi-prone or on their side to ensure safety and prevent aspiration.

- **Weigh the Patient:** If possible, weigh the patient (particularly in children) to calculate accurate medication dosages based on body weight.
- **Rapid Clinical Assessment:** Conduct a swift clinical assessment, paying close attention to signs of meningitis and other potential conditions.
- **Blood Tests:** Collect blood for diagnostic smear/slide or RDT, haematocrit or hemoglobin, and other relevant laboratory tests.
- **Exclude Hypoglycemia:** Check for hypoglycemia and monitor blood glucose levels regularly (every 2-4 hours).
- **Specimen Collection:** Obtain specimens to rule out meningitis or other conditions if clinically indicated.
- **Start Treatment:** Begin treatment for hypoglycemia, meningitis, or any other identified conditions that require immediate attention.
- **Monitor Urine Output:** Closely monitor the patient's urine output; insert a urethral catheter if necessary.
- **Plan Fluid Management:** Administer intravenous fluids during the first 8 hours, including glucose therapy and blood transfusion, if required.
- **Manage Fever:** If the patient's temperature exceeds 38.5°C, take measures to reduce fever.
- **Anticonvulsant Therapy:** Initiate anticonvulsants if convulsions occur and monitor the patient closely throughout the process.

3.4.3 SPECIFIC ANTIMALARIAL TREATMENT

RECOMMENDATION: The initial treatment for all severe malaria cases should be parenteral artesunate

The **MEDICINE OF CHOICE** for the initial treatment of all cases of severe malaria is artesunate, given parenterally either as an intravenous or intramuscular injection. Parenteral treatment should be administered for a minimum of 24 hours and continued until the patient is stable enough to tolerate oral medications, or for a maximum of seven (7) days. Once the patient can tolerate oral medication, complete the treatment with a full 3-day course of the recommended first-line artemisinin-based combination therapy (ACT), which is artemether-lumefantrine (AL).

Box 1: Dosage Schedule for Parenteral Artesunate

Give parenteral artesunate on admission (time=0), 12 and 24 hours, then daily until the patient can tolerate oral medication or up to a maximum of 7 days if the patient is still unable to take oral medication. After 24 hours, when the patient can tolerate oral medication, give the full dose of AL unless the patient has received 7 days of parenteral artesunate.

Children less than 20kg body weight should receive 3.0 mg/kg per dose
Older Children (weighing more than 20kg) and *adults* should receive 2.4 mg/kg per dose.

If artesunate is unavailable, intramuscular artemether should be given for at least 24 hours (irrespective of the patient's ability to tolerate oral medication). Once the patient can tolerate oral medication, complete the treatment with a full 3-day course of artemether-lumefantrine (AL).

The dosage schedule for intramuscular artemether, where artesunate is not available, is provided below:

Box 2: Dosage Schedule for Intramuscular Artemether

Give IM Artemether at 3.2 mg/kg (loading dose) followed by 1.6 mg/kg daily until the patient tolerates oral medication up to a maximum of 7 days if the patient is still unable to take oral medication

3.4.4 ADJUVANT THERAPY FOR SEVERE MALARIA

a) Convulsion

Convulsions are common in children with severe *P. falciparum* but are relatively rare in adults. The general principles for the care of patients with convulsions should be followed: maintenance of a clear airway urgently, aborting the seizure with an anticonvulsant, monitoring of vital signs, and nursing the patient in a recovery position.

Treatment: Treat with diazepam, phenobarbitone, or phenytoin IV.

Anti-convulsant Doses:

Diazepam IV: 0.05ml/kg or 0.25mg/kg

Rectal Diazepam: 0.1ml/kg or 0.5mg/kg

Phenobarbitone loading dose: 15mg/kg IV/IM over 15mins

Phenobarbitone maintenance: 5mg/kg OD to start after 24hrs for 2 days

Phenytoin IV: 15mg/kg over 1 hour

If IV access is not possible, the rectal route should be used. Give rectal diazepam at a dose of 0.5-1.0mg/kg up to a maximum of 2 doses of diazepam; if the convulsion continues, switch to phenobarbitone

NB: for children rectal administration of diazepam is preferable.

Diazepam should not be used in infants below one month of age; instead, use phenobarbitone 20mg/kg IM or IV. If convulsions persist, repeat phenobarbitone 10mg/kg after 30 minutes.

b) Hypoglycemia

Hypoglycemia is common in severe malaria, often due to anorexia, vomiting, or quinine infusion, and can present with symptoms like restlessness, altered consciousness, or seizures. It is a poor prognostic indicator and requires urgent management. Check blood glucose and treat for hypoglycaemia if <3 mmol/L

Treatment: Administer 50 mL of 50% glucose IV rapidly (for children, use 10% dextrose at 2-3ml/kg. If only 50% glucose is available, dilute it to make a 10% solution. Follow with an IV infusion containing 5-10% dextrose until blood glucose stabilizes or use a nasogastric tube for unconscious patients.

If injectable glucose is unavailable, give 50 mL of a 10% glucose solution made with sugar (4-level teaspoons) and water (200 mls) orally or via a nasogastric tube.

c) Hyperpyrexia

Treatment: Give Paracetamol at 15mg/kg every 6 hours until the fever subsides. In addition, expose and fan to cool body temperature

d) Dehydration

Fluid management is critical in patients with severe malaria. Care should be taken not to over-hydrate the patient. If dehydration is present, it should be managed according to standard guidelines (ETAT+ guidelines, IMNCI).

e) Pulmonary Oedema

Pulmonary edema is commonly caused by overhydration. Check for over-hydration and stop all intravenous fluids. Prop up the patient at an angle of 45° , give oxygen (to maintain saturation $>92\%$), and give a diuretic (Furosemide, 1-2mg /kg of body weight for children and up to a maximum of 40mg by intravenous injection for adults. If there is no response, increase the dose progressively to a maximum of 200mg). Intubate and add positive end-expiratory pressure / continuous positive airway pressure in life-threatening hypoxemia (or in case of severe respiratory distress).

f) Severe Anaemia

Anaemia is a major cause of the high morbidity and mortality associated with malaria. It is especially serious in young children and pregnant women. Patients with signs of severe anaemia (severe pallor, rapid, difficulty breathing, inability to feed, and extreme tiredness) must be admitted to a hospital in an emergency.

Diagnosis of severe anaemia

Hb <5 g/dl or Haematocrit $<15\%$ (in children)

Hb <7 g/dL or Haematocrit $<20\%$ (in adults)

Management of severe anaemia

Give urgent blood transfusion over 3 to 4 hours to patients with severe pallor/anaemia in heart failure (Give 5-10mls of packed cells or 10- 20mls of whole blood/kg body in patients with Hb <5 g/dl or Hb <6 g/dl in-patient with signs of heart failure (dyspnoea, enlarged liver, gallop

rhythm). Where blood is not available give pre-referral treatment and refer urgently to a health facility with blood transfusion services.

Follow-up after discharge: Continue with folic acid and ferrous sulphate and review after 14 days to check haemoglobin or haematocrit level. Encourage patients to sleep under an insecticide-treated net (ITN).

g) Acute Kidney Injury (AKI)

Exclude pre-renal causes of AKI such as dehydration.

Management of AKI

Maintain strict fluid balance if in established renal failure by strictly monitoring fluid input and urine output monitoring aim at urine output >0.5 - 1 ml/kg/hour). If indicated carry out peritoneal dialysis or haemodialysis.

h) Circulatory collapse or Shock

In all severe malaria cases with shock, suspect gram-negative sepsis: Take blood for culture, correct hypovolaemia, and commence broad-spectrum antibiotics

CAUTION: Avoid the following:

- Drugs that increase gastrointestinal bleeding.
- Corticosteroids.
- Other anti-inflammatory agents such as Non-Steroidal Anti-Inflammatory Drugs (NSAIDs).
- Other agents given for cerebral oedema (Urea, Mannitol).

3.4.5 SUPPORTIVE CARE AND MONITORING OF PATIENTS WITH SEVERE MALARIA

Timely provision of supportive care is critical for the survival of patients with severe malaria. Patients with severe malaria require intensive nursing care, ideally in an intensive care unit (ICU) when available. Close clinical observation is essential and should be conducted as frequently as possible. This should include regular monitoring of vital signs, coma score, and urine output to assess the patient's condition and guide treatment adjustments. Frequent reassessment helps ensure timely intervention and improves the chances of recovery. Supportive care and monitoring should include, but is not limited to, the following:

- Level of consciousness
- Fluid intake and speed of infusion of fluids
- Fluid output including urine volume (hourly) and specific gravity. If necessary, insert a urethral catheter. (Oliguria <17 ml/hr in an adult or <0.3 ml/kg/hr in infants and children)
- Blood glucose 4 hourly while the patient is unconscious. If it falls below 2.2 mmol/l (<40 mg/dl), correct for hypoglycemia.
- Core temperature: Check if the temperature remains high or increases despite treatment with malaria treatment.

- Respiratory rate 4 hourly (Increased respiratory rate: < 2 months: 60 or more per minute, 2-11 months: 50 or more per minute, 1 year and above: 40 or more per min. or difficulty in breathing. Give appropriate treatment if patients develop secondary pneumonia.
- Parasitaemia: Parasitaemia commonly remains at the initial level for 12-24 hours even if drugs are fully effective. Check if it continues to remain high after 2-3 days or remains positive for 5 days.
- Haemoglobin (Hb): Check if anaemia is suspected or worsening
- Convulsions. These can recur or develop for the first-time during treatment and may be due to hyperpyrexia, abnormal blood glucose or electrolyte imbalance.
- Abnormal bleeding including from vene-puncture sites
- Uterine contractions and fetal heart rate in pregnant women.

3.4.6 ASSESSMENT OF RECOVERY

The absence of sequelae indicates a good recovery from severe malaria treatment. However, it is essential to assess patients, particularly children, for potential neurological complications from the disease or its treatment, as about 10% of children experience neurological sequelae following recovery from cerebral malaria.

a) Assess:

- I. Vision
- II. Hearing

b) Repeat on 7th and 14th day:

- I. Thick and thin blood films
- II. Haematocrit
- III. Haemoglobin

c) For children recovering from severe anaemia:

- I. Give iron and folic acid for two months
- II. If child has sickle cell disease, give folic acid only
- III. Advice on correct feeding recommendation according to the child's age .
- IV. Give anthelmintics if the child is over one year and has not received a dose of
- V. Albendazole/Mebendazole for the past six months.
- VI. Encourage to sleep under ITNs.

d) If neurological defects occur, refer for physiotherapy

3.5 REFERRAL

RECOMMENDATION: All cases diagnosed as severe malaria should be referred for hospitalization

Patients with severe malaria can deteriorate quickly and should be managed in an appropriate inpatient facility. Immediate referral to a hospital is essential, following recommended pre-referral management. All severe malaria cases should be managed in the highest possible facility with admission facilities and special care treatments as may be required

3.5.1 CRITERIA FOR REFERRAL

The following criteria should be used to refer patients:

- **Severe disease:** The presence of clinical, laboratory or imaging finding(s) for severe disease is an indication for immediate referral if the facility is unable to provide effective in-patient management.
- **Failure to respond to recommended therapy:** If the patient is already being managed in a hospital, persistence or worsening of features of severe disease or disease complication(s) should prompt referral to a higher level of care. The decision depends on the capacity of the referring health facility, the patient's clinical progression, and the feasibility of referral options.

One or more of the following listed below is an indication for referral to a higher level of care especially to a hospital:

- Altered consciousness (confusion, change in behaviour, delirium, coma persisting for over 30 minutes after convulsion).
- Convulsions (fits): more than 1 within 24h.
- Repeated vomiting and inability to retain oral medication, food and fluids.
- Persistent hypothermia
- Severe dehydration
- Persistent hyperpyrexia in children within 24 hours to 48 hours (axillary temperature > 38.5°C)
- Severe anaemia (Extreme pallor).
- Circulatory collapse or shock, (feeble, weak, rapid pulse and cold limbs).
- Acute Renal Injury (failure to pass urine or passing very little quantity of urine)
- Obvious Jaundice (yellowness of the eyes).
- Pregnancy with persistent high-grade fever > 39°C
- Failure to respond to initial treatment within two to three days.
- Severe reaction to antimalarials medicines.
- Severe respiratory distress
- Any other conditions that cannot be managed locally.

3.5.2 MANAGEMENT OF SEVERE MALARIA BEFORE REFERRAL

When severe malaria is identified, prompt treatment should be instituted, and arrangements made for referral.

Remember to:

- Start initial antimalarial treatment and supportive care (first dose of IM artesunate, or rectal artesunate).
- Establish communication early with the intended receiving facility.
- Continue feeding if possible
- Include a referral note about the clinical picture, the type of treatment given, dosages, times and route of administration for any medications given
- Send a staff member with the patient if possible
- Send potential blood donors with the patient if applicable

3.5.3 PRE-REFERRAL ANTIMALARIAL TREATMENT OPTIONS

- Rectal Artesunate (10 mg/kg body weight) **ONLY** in children less than 6 years of age
- Intramuscular Artesunate (3 mg/kg if weight <20 kg, 60mg if weight 20-25 kg, and 2.4 mg/kg for body weight $\geq$ 25 kg)
- When IM Artesunate is not available, give Intramuscular Artemether (3.2 mg/kg body weight as a loading dose)

NB: If, however, referral is impossible or delayed (e.g. due to logistics reasons etc) continue the treatment with rectal Artesunate or IM Artesunate (as applicable) until the patient can tolerate oral medication. At this point, continue with the follow-on treatment recommendation.

Table 8: Dosage schedule for rectal artesunate suppository (100mg)

Age	Dosage
6 months to 3 years	1
3 years to <6 years	2

CHAPTER 5 MALARIA CHEMOPREVENTION AND CHEMOPROPHYLAXIS

Malaria Chemoprevention is the intermittent use of a complete treatment course of antimalarials to prevent malaria in targeted vulnerable populations, while chemoprophylaxis is the regular use of anti-malarial drugs to prevent the development of malarial parasitemia.

Current chemopreventive/chemoprophylactic interventions in Sierra Leone include:

- i. Intermittent Preventive Treatment of Malaria in Pregnancy
- ii. Perennial Malaria Chemoprevention
- iii. Chemoprophylaxis for non-immune travelers

4.1 INTERMITTENT PREVENTIVE TREATMENT IN PREGNANCY

Intermittent Preventive Treatment in Pregnancy (IPTp) is the presumptive provision of a full treatment course of an efficacious antimalarial at specific intervals during pregnancy. IPTp has

been shown to reduce the risk of placental infection and the associated risk of maternal anemia, miscarriage, premature deliveries, and low birth weights. Sierra Leone recommends that in areas of stable transmission, all pregnant women should receive at least three doses or more of IPTp as early as possible in the second trimester (13-16 weeks) during routine scheduled antenatal visits. Each treatment should be given at least 4 weeks apart. The current recommended medicine for IPTp is three tablets of Sulphadoxine- Pyrimethamine (SP)¹.

4.2 PERENNIAL MALARIA CHEMOPREVENTION

Perennial Malaria Chemoprevention (PMC) is the administration of anti-malarial to children at the highest risk of malaria at specific ages throughout the year. In Sierra Leone, it is given at 10 weeks, 14 weeks, and 9 months, which aligns with the EPI schedule of Penta2, Penta3, and Measles/Yellow fever vaccinations. The drug of choice for PMC is Sulphadoxine-Pyrimethamine (SP).

4.3 MALARIA CHEMOPROPHYLAXIS

Malaria prophylaxis is not necessary for persons living in a malarious area because it may lower one's resistance to the disease. However, it may be used in non-immune visitors because of the risk of severe disease. It is, however, not 100% protective, and non-immune visitors should also use other preventive measures, such as sleeping under LLINs and reducing time spent outdoors after dusk.

4.3.1 CHEMOPROPHYLAXIS OPTIONS FOR NON-IMMUNE VISITORS

The options available for chemoprophylaxis include mefloquine, atovaquone-proguanil (Malarone), and doxycycline. The choice from these options is determined by tolerability, as all these options are efficacious in Sierra Leone.

- **Mefloquine (Lariam)** should be started at least 2 weeks before the arrival to ensure that the individual can tolerate and has achieved the desired blood levels of the drug before arrival in a malaria endemic area. Also, it should be continued for four weeks after departure and return to the country of origin. Mefloquine is dosed weekly.
- **Atovaquone-proguanil** can be started 24 hours before arrival and stopped one week after return. It is, however, dosed daily.

Table 9: Dosage schedule for Malaria Chemoprophylaxis among Non-Immune Individuals

Medicine	Weight	Dosage	Comments
Mefloquine	Children	5mg/kg once weekly	-Safe in Pregnancy and breastfeeding mothers -Not recommended for children <5kg
	Adults	250mg once weekly	-Not to be given with Oral Typhoid vaccine

¹ Please refer to the Malaria in Pregnancy guidelines for additional information such as the dosing schedule and the community component on IPTp

Atovaquone-proguanil	11-20kg	62.5 mg atovaquone/25mg proguanil daily	Not recommended in Pregnancy and breastfeeding mothers
	21 – 30kg	125 mg atovaquone/50mg proguanil daily	-Not recommended in children <11kg -Take with food or milky products
	31-40kg	187.5 mg atovaquone plus 75mg proguanil	-May interfere with Rifampicin, Rifabutin, Metoclopramide, Tetracycline, Live Typhoid vaccine
	>40kg	250mg atovaquone/100mg proguanil daily	

CHAPTER 6 SOCIAL BEHAVIOUR CHANGE

Effective social and behavioural change (SBC) for malaria elimination requires alignment with the National Malaria Control Behaviour Change Communication Strategy and approval from relevant authorities. Efforts should target both health workers and the general public, using multiple communication channels across public and private sectors. Clear messaging about potential adverse drug reactions (ADRs) can build trust in malaria treatments, while interpersonal communication is key to promoting positive health behaviors. This comprehensive approach aims to foster social norms that support malaria prevention, diagnosis, and treatment for sustainable elimination outcomes.

Healthcare and community-based health workers should serve as key agents in implementing malaria prevention and treatment protocols at care points and within communities. They should utilize every opportunity to educate patients and the public on malaria prevention and treatment, promoting awareness to support elimination efforts.

Healthcare and community-based health workers should actively promote malaria interventions by raising awareness of their benefits for individuals and communities, as well as the risks of non-compliance. To enhance communication, they should use materials like job aids, counseling cards, posters, flip charts, and leaflets for effective engagement with patients and communities

5.1 KEY MESSAGES

Key messages the Health Worker should stress to patients and the general public include the following:

For any case of fever:

- Visit the nearest health facility as soon as possible to seek prompt care within 24 hours.
- Get tested to confirm malaria and follow the prescribed antimalarial treatment from a health worker. If the test is negative, ask to be checked for other possible conditions.
- Do not take ACTs if you test negative for malaria.
- Always test first, as many illnesses can cause fever similar to malaria. Use a malaria rapid diagnostic test (RDT) or microscopy for accurate diagnosis.
- Do not advise others to use the same ACT prescribed to you; instead, encourage them to visit a health facility for proper testing and treatment.
- Follow the treatment exactly as prescribed, without stopping early or sharing ACTs.
- Report any severe symptoms, lack of response to treatment, or adverse reactions to the nearest health facility immediately.
- Pre-referral treatment for severe malaria is not complete treatment and must be followed by referral to a higher-level of care where complete treatment is available.
- Treating malaria fully also contributes to the reduction in malaria transmission.

Note: Always look for other signs of fever if even the malaria test is positive

Other key malaria messages

- Early attendance by pregnant women at antenatal clinics to receive full course of IPTp.
- Encourage caregivers to take their eligible children for malaria vaccine and PMC during immunization.
- Promotion of prompt and effective treatment of malaria by community health workers (CHWs).
- Regular and consistent use of vector control tools to protect households from malaria (ITNs, window screens, environmental sanitation, etc.,).

Information for Special groups

- Malaria can exacerbate symptoms of sickle cell disease and therefore this population will benefit from malaria preventive services such as ITN, SBC and prompt access to health services for treatment.
- Health workers should have a high index of suspicion of malaria in children with severe malnutrition because they have high mortality from malaria and presenting symptoms may be atypical as signs of severity maybe masked

Mothers and other Caregivers in our communities play a key role in malaria case management in the home where the first step in managing the disease very often starts. Health care workers must especially target this sub-population

CHAPTER 7 PHARMACOVIGILANCE

Pharmacovigilance is defined as the science and activities related to detecting, assessing, understanding, and preventing adverse effects or any other drug-related problems. It aims to ensure that medications, once they are in widespread use, remain safe and effective for the public by monitoring for any issues that may not have been identified during clinical trials.

6.1 WHEN TO REPORT

Adverse Drug Reaction (ADR) reporting is crucial for patient safety and can be categorized as either expedited or non-expedited based on the severity and nature of the reaction.

Expedited ADR Reporting: Expedited reports cover ADRs classified as serious (both expected and unexpected). An *expedited* ADR must be reported quickly—ideally immediately, but no later than 7 calendar days from the date of receipt of the report. Serious adverse events generally include any reaction that leads to:

- Death
- Hospitalization or prolongation of an existing hospitalization
- Significant disability or incapacity
- Congenital malformation
- Life-threatening conditions

Non-Expedited ADR Reporting: For ADRs considered *non-serious*, reporting is still required but with a less urgent timeline, allowing up to 28 days from the time the event is reported. Non-serious ADRs, while less critical, still contribute valuable information for assessing a drug's overall safety profile.

Ensuring timely reporting of both types of ADRs helps maintain drug safety standards and protect public health.

6.2 HOW TO REPORT

- Use the standard ADR form (see Annex G)
- Med Safety App
- Telephone call (update to latest)

Submission Process: Completed paper forms should be submitted to the DHMT. From there, they will be forwarded to reach the appropriate authorities in Sierra Leone.

6.3 THERAPEUTIC EFFICACY TESTING OF ANTIMALARIALS

Therapeutic Efficacy Testing (TET) involves monitoring the efficacy of anti-malarial treatments regularly to combat drug resistance, following a standard protocol. Regular malaria efficacy testing provides valuable data for updating and refining treatment guidelines. It helps determine which anti-malarial drugs are most effective in specific regions and against certain strains of the malaria parasite.

CHAPTER 8 MONITORING AND EVALUATION

Monitoring and Evaluation (M&E) is essential for tracking and guiding the implementation of malaria elimination interventions and assessing progress toward goals. Key data sources include routine surveillance, operational research, and surveys. Malaria case management data will primarily be tracked through the District Health Information System (DHIS), while the Malaria Indicator Survey (MIS), Demographic and Health Survey (DHS), and Multiple Indicator Cluster Survey (MICS) will evaluate program outcomes and impact. Specific indicators for malaria case management will also be monitored by the program.

A detailed table with malaria indicators is described in Monitoring and Evaluation Plan of the National Malaria Strategic Plan.

Regular collection, analysis and use of malaria information to inform programming and decision-making within the NMCP.

7.2 DATA COLLECTION AND REPORTING

Malaria surveillance in Sierra Leone is integrated into the District Health Information System (DHIS).

- At the health facility level, the malaria patient information is completed in the facility Register (one for under five years old and another for above 5 years old).

- Monthly, data will be aggregated in summary forms (PHU-HF1, PHU-HF2, PHU-HF3, PHU-HF4, HF5, HF6).
- The summary forms are sent to Chiefdom level.
- Chiefdom level staff will enter the data into DHIS monthly.

Note: NMCP has access to DHIS to review malaria indicators

Health workers should ensure that all appropriate forms are completed and sent in a timely manner at all levels according to national reporting timelines.

7.3 DATA ANALYSIS AT HEALTH FACILITY

To monitor the trend of cases at health facility catchment area, the number of cases should be filled in the weekly malaria epidemiological chart. It is recommended that case data is reviewed on a weekly basis.

REFERENCES

National Health Sector Strategic Plan (NHSSP 2021-2025)

World Health Organization (WHO) Guidelines on Malaria (October 2023)

Therapeutic Efficacy Studies (TES) in Sierra Leone

Malaria Indicator Survey (MIS)

Demographic and Health Survey (DHS)

Multiple Indicator Cluster Survey (MICS)

Sierra Leone National Malaria Elimination Strategic Plan (2021–2025)

Emergency Triage, Assessment, and Treatment Plus (ETAT+) Guidelines

Integrated Management of Neonatal and Childhood Illnesses (IMNCI)

Glasgow Coma Scale (GCS)

Blantyre Coma Scale

Sierra Leone Basic Package of Essential Health Services (2015)

WHO Recommendations for Monitoring Antimalarial Drug Resistance

Sierra Leone Suspected Adverse Drug Reaction (ADR) Reporting Form

Global Malaria Technical Strategy (2021–2030)

WHO Guidelines for Intermittent Preventive Treatment in Pregnancy (IPTp)

WHO Guidelines on Malaria Chemoprophylaxis

WHO Recommendations for Pre-Referral Rectal Artesunate Use

Sierra Leone National Pharmacovigilance Guidelines

ANNEXES

Annex A: Reporting of Blood Smear Results

Two methods are commonly used in reporting blood smear results.

a. Thick film:

In this method it is assumed that 1 microlitre (ul) of blood contains 8,000 white blood cells (WBC). The number of parasites counted relative to the number of leucocytes counted can thus be converted to the number of parasites per ul of blood by the simple formula given below:

$$\frac{\text{Number of parasites} \times 8000 \text{ WBC/ul}}{\text{Number of leucocytes counted}} = \text{Parasite count per ul}$$

b. Thin film

$$\frac{\text{Number of parasitized red cells} \times 5\,000\,000}{\text{Number of red cells counted}} = \text{Parasite count per ul}$$

Number of leucocytes counted

Annex B: Supportive Care/ Monitoring of Patients with Severe Malaria

Patients with severe malaria required intensive nursing care, preferably in an intensive care unit where possible. Clinical observation should be made as frequently as possible. This should include monitoring of vital signs, coma score, urine output.

The following should be monitored:

1. Level of consciousness Fluid intake/output and speed of infusion of fluids.
2. Urine volume (hourly) and specific gravity. If necessary, insert a urethral catheter. (Oliguria<17ml/hr in an adult or <0.3ml/kg/hr in infants and children)
3. Blood glucose 4-hourly while the patient is unconscious. If it falls below 2.2 mmol/l (<40 mg/dl), correct for hypoglycemia.
4. Blood pressure (<90mm hg systolic in an adult, <50mm Hg in infants and children)
5. Core temperature(>-38.5°C): If temperature remains high or increases despite treatment with malaria treatment
6. Respiratory rate 4-hourly (Increased respiratory rate: <2months: 60 or more per minute, 2-11 months: 50 or more per minute, 1 year and above: 40 or more per min. or difficulty in breathing.
7. Patients with secondary pneumonia or with clear evidence of aspiration should be given appropriate treatment.
8. Parasitaemia: Remains high 2-3 days or remains positive for 5 days. Parasitaemia commonly remains at the initial level for 12-24 hours, even if drugs are fully effective.
9. Haemoglobin (Hb) if anaemia is suspected to be worsening.: Falls below 5g/dl or Haematocrit <15%
10. Occurrence of convulsions. These can recur or develop for the first-time during treatment and may be due to hyperpyrexia, abnormal blood glucose, or electrolyte imbalance.
11. Bleeding from venepuncture sites or spontaneous haemorrhage
12. Uterine contractions and fetal heart rate in pregnant women.

Annex C: The Glasgow Coma Scale (For Adults and Children >5 Years)

To obtain the Glasgow coma score, obtain the score for each criterion, then add the three figures to obtain a total.

Table 10: Glasgow coma score

Criteria	Response	Score
Eyes opening (4)	Spontaneously	4
	To speech (verbal stimuli)	3
	To pain only	2
	No response	1
Best verbal response (5)	Oriented	5
	Confused	4
	Inappropriate words	3
	Incomprehensible sounds	2
	None	1
Best motor response (6)	Obeys commands	6
	Localizes painful stimulus	5
	Withdraws in response to pain	4
	Flexion in response to pain	3
	Extension in response to pain	2
	None	1
Total		3-15

Annex D: The Blantyre Coma Scale

This score has been modified to be applicable to children, including those who have not learned to speak

Criteria	Response	Score
Eye movements (1)	Directed (e.g. Follow mothers)	1
	Not directed	0
Verbal response (2)	Appropriate cry	2
	Moan or inappropriate cry	1
	None	0
Best motor response (2)	Localizes pain stimulus	2
	Withdraws limb from pain	1
	Nonspecific or absent response	0
Total		0-5

To elicit pain during coma evaluation:

- (a) Rub knuckles on patient's sternum.
- (b) Firm pressure on thumb nail with horizontal pencil.

A state of unrousable coma is reached at a score <3. Scoring can be repeated to assess improvement or deterioration.

Annex E: Integrated Management of Childhood Illness

ASK THE MOTHER WHAT THE CHILD'S PROBLEMS ARE

- Determine whether this is an initial or follow-up visit for this problem.
- if follow-up visit, use the follow-up instructions on *TREAT THE CHILD* chart
- if initial visit, assess the child as follows:

USE ALL BOXES THAT MATCH THE
CHILD'S SYMPTOMS AND PROBLEMS

CHECK FOR GENERAL DANGER SIGNS

ASSESS

ASK	LOOK
• Is the child able to drink or breastfeed?	• See if the child lethargic or unconscious?
• Does the child vomit everything?	• Is the child convulsing now?
• Has the child had convulsions?	

URGENT
ATTENTION

SIGNS

- Any general danger sign

CLASSIFY AS

VERY SEVERE
DISEASE

IDENTIFY TREATMENT

- Give diazepam if convulsing now
- Quickly complete the assessment
- Give first dose of IM Ampicillin and IM Gentamycin
- Treat to prevent low blood sugar
- Keep the child warm
- Refer urgently

A child with any general danger sign needs **URGENT** attention; complete the assessment and give any pre-referral treatment immediately so that referral is not delayed.

* If the child is convulsing now, manage the airway and treat the child with Diazepam rectally. Then immediately assess, classify and provide other treatment before urgently referring to hospital.

Does the child have fever?

(by history or feels hot or temperature 37.5°C* or above)

IF YES:

Has the child travelled to a high risk (Malaria endemic, seasonal transmission or epidemic prone) area in the last 1 month?

Decide Malaria Risk: high or low risk.

THEN ASK:

- For how long?
- If more than 7 days, has fever been present every day?
- Has the child had signs of measles within the last 3 months?

LOOK AND FEEL:

- Look or feel for stiff neck.
- Look for runny nose.

Look for signs of MEASLES:
• Generalized rash and one of these: cough, runny nose, or red eyes.

- Look for any other cause of fever***

CLASSIFY
FEVER: HIGH OR
LOW MALARIA
RISK

High Malaria risk:

Do a malaria test.

- Endemic Zone
- Seasonal Transmission Zone
- Epidemic prone areas

Low malaria risk:

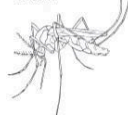
- Do a malaria test if no obvious cause of fever

NOTE: If you can't test, don't withhold treatment

TEST POSITIVE
• *P.falciparum* PRESENT
• *P.vivax* PRESENT

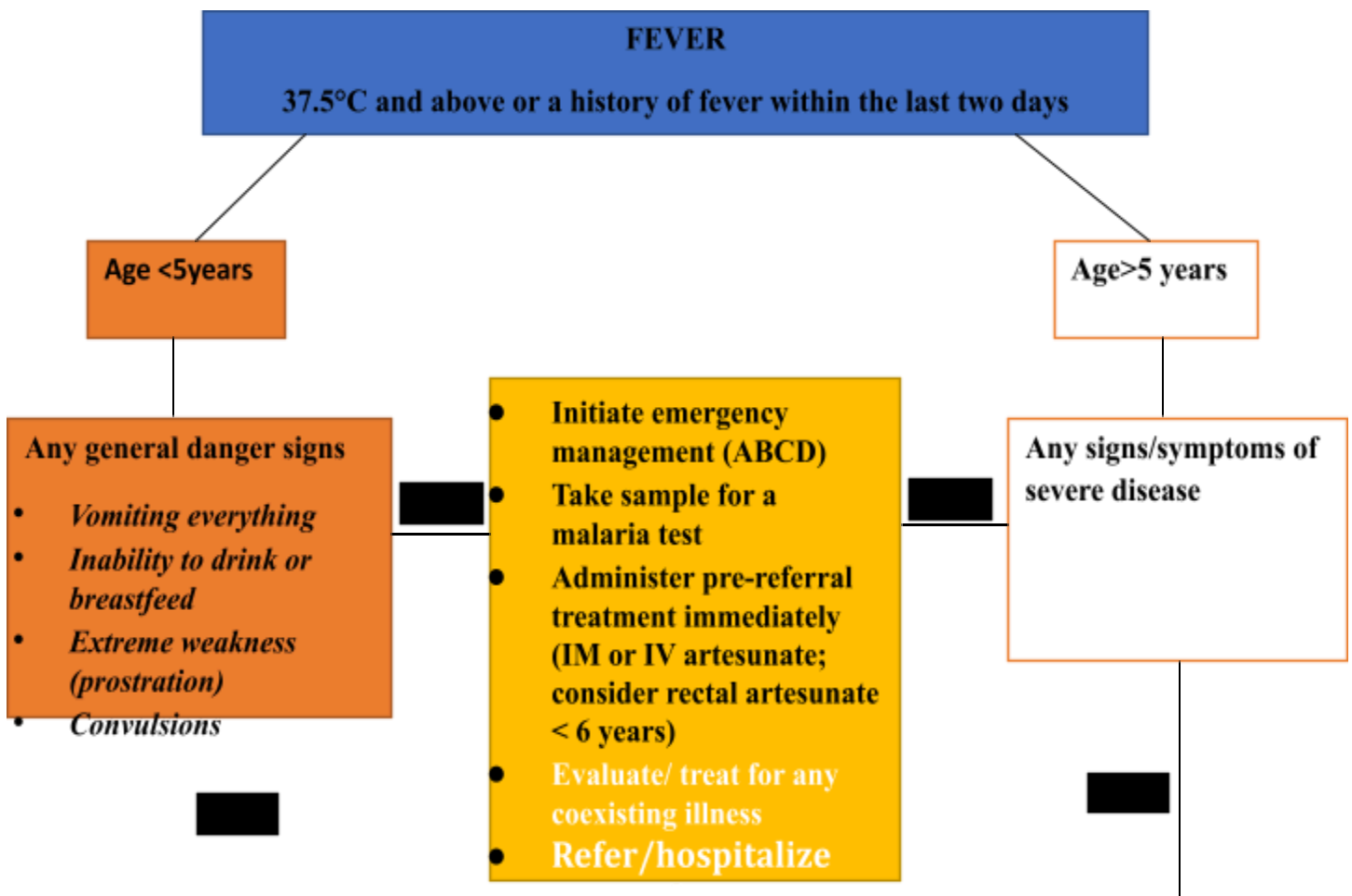
TEST NEGATIVE

- *P.falciparum* or *P.vivax* absent



CLASSIFY		IDENTIFY TREATMENT
- Any general danger sign or Stiff neck. - AND - Confirm malaria with a test.	VERY SEVERE FEBRILE DISEASE OR SEVERE MALARIA	- Give first dose of Artesunate for severe malaria (see pg 16) - Give first dose of Ceftriaxone (see pg 16) - Treat the child to prevent low blood sugar (see pg 17) - Give one dose of paracetamol in clinic for high fever ($\geq 38.5^{\circ}\text{C}$) (see pg 14) - Admit or Refer URGENTLY to hospital - Screen for possible TB disease and check for HIV.
Malaria test POSITIVE**	UNCOMPLICATED MALARIA	- Give Artemether + Lumefantrine (AL) (see pg 16) - Give one dose of paracetamol in clinic for high fever ($\geq 38.5^{\circ}\text{C}$) (see pg 14) - Give Vitamin A (see pg 14) - Follow up in 3 days if fever persists (see pg 23) - If fever is present every day > 7 days assess further or refer - Screen for possible TB disease and check for HIV - Advise when to return immediately.
- Malaria test NEGATIVE - Runny nose PRESENT - Measles PRESENT or - Other causes of fever PRESENT	FEVER: NO MALARIA	- Give one dose of paracetamol in clinic for high fever ($\geq 38.5^{\circ}\text{C}$) - Assess for other possible causes of fever - Follow up in 3 days if fever persists (see pg 23) - If fever is present every day for more than 7 days assess further or refer (see pg 23) - Screen for possible TB disease and check for HIV - Advise mother when to return immediately.

Annex SEQ Annex * ALPHABETIC F: Flow Chart for Diagnosis and Management of Malaria at the Periphery



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Annex G: Sierra Leone Suspected Adverse Drug Reaction Reporting Form

Figure 4: Suspected Adverse Drug Reaction Reporting Form (Page 1)

 Drug Safety Monitoring Programme Pharmacy Board of Sierra Leone Ministry of Health And Sanitation						No:
Suspected Adverse Drug Reaction Reporting Form						
1. Patient Details						
Last Name :			Sex:			
Other Name (s) :			Age (Years):			
Address :						
Inpatient/Outpatient Number :			Weight (kg):			
Hospital/Treatment Centre:						
2. Suspected Drugs/Products						
Brand Name:	Strength	Generic Name:	Batch No (if any)	Man. Date	Expiry Date	
3. Name and Address of Manufacturer						
Therapeutic Indication (Reason for Use)		Daily dose	Route	Date started	Date stopped	
4. Drugs taken Concomitantly			All concomitant drugs including herbal preparations			
Brand or Generic Name	Daily dosage	Route	Date started	Date stopped	Reasons for use	
COMMENTS: (e.g. Relevant history, Allergies, Previous exposure, Baseline test results/laboratory data). Add separate sheet if necessary.						
5. Source of Drug						
Prescribed? Yes ☐ No ☐		Obtained over the counter? ☐ Yes ☐ No ☐				
Hospital Pharmacy/Health Center ☐	Community Pharmacy OR Drug Store ☐	Patent Medicine Shop ☐	Herbalist/Traditional Healer ☐	Relative /Friend / street vendor ☐	Other (Specify) ☐	
6. Details of reaction experienced by the patient (Please Describe)				(Use separate sheet if necessary)		
Date / time reaction started:		Date / Time reaction stopped:	Was patient admitted: Yes No ☐		Duration of admission (hours)	

Figure 5: Suspected Adverse Drug Reaction Reporting Form (Page 2)

7. Adverse Reaction Outcome		(Tick all that apply)	
☐ Death	☐ Life threatening	Event reappeared on rechallenge: Yes ☐ No ☐	Recovered Yes ☐ No ☐
☐ Disability	☐ Hospitalization	Rechallenge not done: Yes ☐ No ☐	Recovered With sequelae Yes ☐ No ☐
☐ Congenital anomaly	☐ Other	Treatment (of Reaction).....	Describe Sequelae.....
			Reaction Continuing Yes ☐ No ☐
8. Product Quality Complaints			
Brand or Generic Name	Batch No.	Dosage form & Strength	Mfg. date
			Expiry Date
			Pack Size
			Type of Container
9. Causality Assessment			
Certain ☐	Probably/Likely ☐	Possible ☐	Unlikely ☐ Unclassified ☐
10. Health Care Professional/Reporter Detail			
Doctor ☐	Pharmacist ☐	Nurse ☐	Other ☐ (Specify)
	Pharm. Tech ☐	CHO ☐	
Name:	Fax:		
Address:	Telephone:	Mobile:	
Email:	Date:	Signature:	

For all questions relating to actual or suspected Adverse Drug Reactions, please call The Pharmacy Board, Drug Information And Pharmacovigilance Unit during working hours on 225983 / 228497 or email us on info@pharm_pbst@yahoo.com. Please return this form to The Pharmacy Board of Sierra Leone 64 Siaka Stevens Street Freetown PMB 322. Fax 224526, OR to any of the Regional Offices in BO, Mobile: 076-680-321, KENEMA, Mobile: 076-680-966, KONO, Mobile: 076-713-946, MAKENI, Mobile: 076-697-227 OR KAILAHUN, Mobile: 076-685-690
For further information please visit our website at www.pharmboard.org

(Please note this report does not constitute an admission that the reporting medical professional or the suspected product caused or contributed to the event)

ADVICE ABOUT VOLUNTARY REPORTING

Report adverse experiences with: ❖ Medications (Drugs and Biologicals) ❖ Medical devices (including in-vitro diagnostics) ❖ Traditional and herbal remedies ❖ Cosmetics ❖ Nutritional Agents	Report Product Quality Problems such as: ❖ Suspected contamination ❖ Questionable stability ❖ Defective components ❖ Poor packaging or labelling ❖ Therapeutic failure
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Report even if:

- ❖ You're not certain the product caused the event
- ❖ You don't have all the details

Confidentiality: Identities of the reporter and patient will remain strictly confidential.

Your support of the Drug Safety Monitoring Programme is much appreciated. Information provided by you will contribute to the improvement of drug therapy in Sierra Leone.